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GitHub Link	https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO4

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

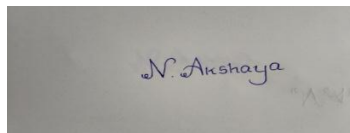
Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I **N.AKSHAYA(192425129)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: N.AKSHAYA

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Total						

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

Answer:

- **Title:** Hierarchical image classification using transfer learning to improve deep learning model performance for Amazon parrots
- **Authors:** Jung-Il Kim, Jong-Won Baek, Chang-Bae Kim
- **Journal:** Scientific Reports
- **Year:** 2025
- **DOI:** 10.1038/s41598-025-88103-3
- **Technique:** Transfer Learning with YOLOv5s
- **Application:** Amazon parrot species classification

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

Answer:

- To accurately detect and classify Amazon parrot species from images.
- The problem is important for wildlife monitoring and conservation.
- Similar morphological characteristics make species classification difficult.
- Existing approaches face limitations due to limited and imbalanced wildlife datasets.

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

Answer:

- To construct a hierarchical classification system for Amazon parrots.
- To apply transfer learning between classification levels.
- To improve species classification accuracy.
- To compare hierarchical and non-hierarchical models.
- To support automated wildlife monitoring.

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

Answer:

- **Data type:** Wildlife images.
- **Species:** 35 Amazon parrot species.
- **Sources:** eBird, iNaturalist and Google Images.
- Images were filtered and manually verified.
- Images were annotated using bounding boxes.
- **Training:** 64%
- **Validation:** 16%
- **Testing:** 20%
- Image size was standardized to **640 × 640**.

5. Proposed Methodology

Explain the overall methodology adopted in the article.

Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

Answer:

- Collect Amazon parrot images.
- Filter and verify images.
- Annotate parrot objects using bounding boxes.
- Resize images to 640 × 640.
- Apply data augmentation.
- Train a class-level YOLOv5s model.
- Transfer learned features to subclass-level classification.
- Transfer learned features to species-level classification.
- Evaluate the final model using precision, recall and MAP.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

Answer:

- The study uses **Transfer Learning**.
- YOLOv5s is initially trained at the class level.
- Learned knowledge is transferred to the subclass-level model.
- The subclass model is further transferred to the species-level model.
- This reduces the difficulty of learning species-level features from limited data.

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers →

Classification/Decoder → Output

Identify the role of the major components in the architecture.

Answer:

Input Image → Preprocessing → Data Augmentation → YOLOv5s Feature Extraction → Class-Level Model → Transfer Learning → Subclass-Level Model → Transfer Learning → Species-Level Model → Detection & Classification Output

- **Input:** Amazon parrot image.
- **Preprocessing:** Resizing and augmentation.
- **YOLOv5s:** Object detection and classification.
- **Transfer Learning:** Reuses learned features between hierarchical levels.
- **Output:** Bounding box and predicted species.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score , Loss , Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

Answer:

Metric	Non-Hierarchical	Hierarchical
Precision	0.890	0.904
Recall	0.855	0.899
mAP@0.5	0.908	0.944
mAP@0.5–0.95	0.785	0.835

9. Results and Discussion

What are the major findings of the research?

How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?

What factors contributed to the observed performance?

Answer:

- The hierarchical transfer-learning model performed better than the non-hierarchical model.
- Precision increased from **0.890 to 0.904**.
- Recall increased from **0.855 to 0.899**.
- mAP@0.5 increased from **0.908 to 0.944**.
- mAP@0.5–0.95 increased from **0.785 to 0.835**.
- The approach improved classification of morphologically similar species.

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

Answer:

- Improved classification accuracy.
- Effective feature reuse through transfer learning.
- Better recognition of visually similar species.
- Reduces manual wildlife identification effort.
- Useful for wildlife monitoring and conservation.
- Provides reliable object-detection performance.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting, Limited generalization, Model complexity.

Answer:

- Dataset imbalance between species.
- Some species have very few training images.
- Some species still show low classification performance.
- High computational requirements.
- Training requires considerable time.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance?

Answer:

- YOLOv5s is appropriate because the task requires object detection and classification.
- The hierarchical architecture is suitable for visually similar species.
- The dataset contains 35 species but has class-imbalance issues.
- Precision, recall and mAP are appropriate evaluation metrics.
- The experimental results are convincing because the hierarchical model outperformed the baseline.
- Newer architectures such as YOLOv9, YOLOv10 or Vision Transformers could be investigated.

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

Answer:

- To increase the number of images for underrepresented species.
- To use GANs for generating additional training images.
- To improve dataset balance.
- To evaluate newer YOLO architectures.
- To investigate advanced transfer-learning methods.
- To deploy the model in automated wildlife monitoring systems.
- To integrate the model with camera-trap systems.

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

Answer:

The article proposes a hierarchical transfer-learning approach using **YOLOv5s** for Amazon parrot classification. The model transfers knowledge from class-level to subclass-level and species-level classification. The hierarchical model achieved **0.944 mAP@0.5**, outperforming the non-hierarchical model with **0.908 mAP@0.5**. The approach is effective for wildlife classification, although dataset imbalance and limited performance for some species remain challenges.

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

Answer:

1. Kim, J.-I., Baek, J.-W., & Kim, C.-B. (2025). Hierarchical image classification using transfer learning to improve deep learning model performance for Amazon parrots. *Scientific Reports*, 15, 3790. DOI: 10.1038/s41598-025-88103-3.
2. Kim, J.-I., Baek, J.-W., & Kim, C.-B. (2022). Image classification of Amazon parrots by deep learning: a potentially useful tool for wildlife conservation. *Biology*, 11, 1303.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis